



# 18<sup>th</sup> International Conference on **Parallel Problem Solving From Nature**

September 14 - 18, 2024  
Hagenberg, Austria

**Griaß eich!** [gri:əs aɪç] *common Austrian greeting formula*

Welcome to the International Conference on Parallel Problem Solving From Nature, held in the Softwarepark Hagenberg, nestled in the picturesque Mühlviertel region of Upper Austria. Here, amidst rolling hills and lush landscapes, we blend modern software technology and cutting-edge AI with the warm hospitality of our beautiful region. We look forward to an inspiring gathering filled with insightful discussions and a dash of Upper Austrian Gemütlichkeit. Let's innovate, collaborate, and maybe even enjoy a slice of local Linzer Torte and a sip of Mühlviertler beer!

**Welcome to Hagenberg!**

*PPSN 2024 is hosted and supported by*



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# Program at a Glance

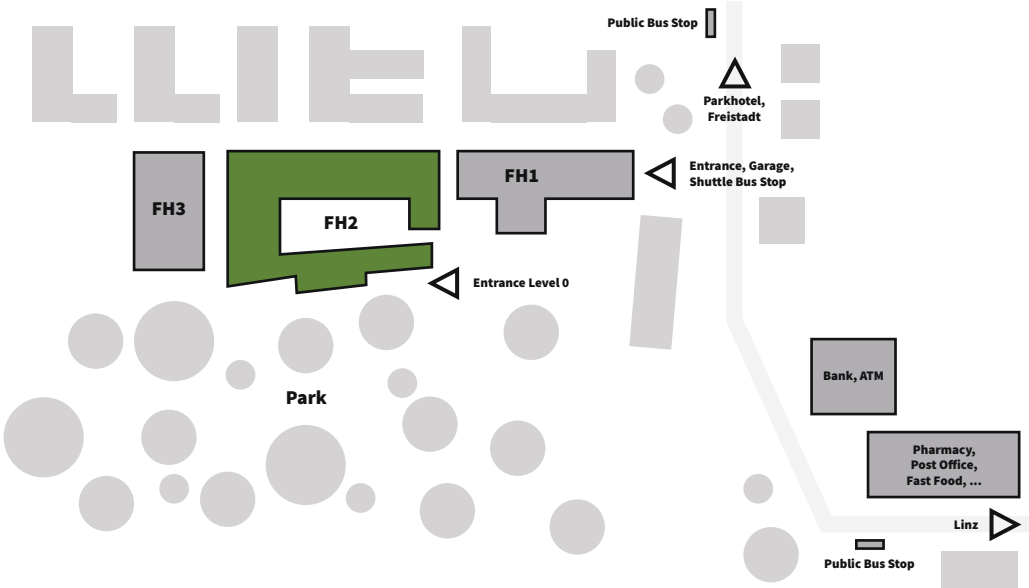
You can find the latest information on our website at [ppsn2024.fh-ooe.at/program](https://ppsn2024.fh-ooe.at/program)

|       | Saturday, 14.9.2024                                                   |                                                                                     |                                                                                                                                          | Sunday, 15.9.2024                                                                         |                                                                            |                                                                                                            |
|-------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| 8:00  | Registratgion & Info Desk<br>8:00 - 9:00                              |                                                                                     |                                                                                                                                          | Registratgion & Info Desk<br>8:00 - 9:00                                                  |                                                                            |                                                                                                            |
| 8:30  | Lecture Hall Corridor                                                 |                                                                                     |                                                                                                                                          | Lecture Hall Corridor                                                                     |                                                                            |                                                                                                            |
| 9:00  | <b>Tutorial 1</b><br>9:00 - 10:30                                     | <b>Tutorial 2</b><br>9:00 - 10:30                                                   | <b>Workshop 1</b><br><br>9:30 - 12:30<br><b>Audimax</b><br><i>BOSS: Bayesian and Surrogate-assisted Search and Optimisation</i>          | <b>Tutorial 9</b><br>9:00 - 10:30                                                         | <b>Tutorial 10</b><br>9:00 - 10:30                                         | <b>Workshop 3</b><br><br>9:30 - 12:30<br><b>Audimax</b><br><i>30 Years of EDAs</i>                         |
| 9:30  | HS3<br><i>Decomposition Evolutionary Multi-Objective Optimization</i> | HS4<br><i>A Gentle Introduction to Theory (for Non-Theoreticians)</i>               |                                                                                                                                          | HS3<br><i>Pareto Optimization for Subset Selection: Theories and Practical Algorithms</i> | HS4<br><i>Advanced Use of Automatic Algorithm Configuration</i>            |                                                                                                            |
| 10:00 |                                                                       |                                                                                     |                                                                                                                                          |                                                                                           |                                                                            |                                                                                                            |
| 10:30 | <b>Coffee Break</b><br>10:30 - 11:00, Break Room                      |                                                                                     |                                                                                                                                          | <b>Coffee Break</b><br>10:30 - 11:00, Break Room                                          |                                                                            |                                                                                                            |
| 11:00 | <b>Tutorial 3</b><br>9:00 - 10:30                                     | <b>Tutorial 4</b><br>9:00 - 10:30                                                   |                                                                                                                                          | <b>Tutorial 11</b><br>9:00 - 10:30                                                        | <b>Tutorial 12</b><br>9:00 - 10:30                                         |                                                                                                            |
| 11:30 | HS3<br><i>CMA-ES</i>                                                  | HS4<br><i>Mathematical Programming as a Complement to Bio-Inspired Optimization</i> |                                                                                                                                          | HS3<br><i>Introduction to Evolution Strategies for Constrained Optimization Problems</i>  | HS4<br><i>Runtime Analysis of Population-based Evolutionary Algorithms</i> |                                                                                                            |
| 12:00 |                                                                       |                                                                                     |                                                                                                                                          |                                                                                           |                                                                            |                                                                                                            |
| 12:30 | <b>Lunch Break</b><br>12:30 - 13:30<br>Break Room                     |                                                                                     |                                                                                                                                          | <b>Lunch Break</b><br>12:30 - 13:30<br>Break Room                                         |                                                                            |                                                                                                            |
| 13:00 |                                                                       |                                                                                     |                                                                                                                                          |                                                                                           |                                                                            |                                                                                                            |
| 13:30 | <b>Tutorial 5</b><br>13:30 - 15:00                                    | <b>Tutorial 6</b><br>13:30 - 15:00                                                  | <b>Workshop 2</b><br><br>13:30 - 17:00<br><b>Audimax</b><br><i>Good Benchmarking Practices for Evolutionary Computation BENCHMARKING</i> | <b>Tutorial 13</b><br>13:30 - 15:00                                                       | <b>Tutorial 14</b><br>13:30 - 15:00                                        | <b>Workshop 4</b><br><br>13:30 - 17:00<br><b>Audimax</b><br><i>Multimodal Multi-objective Optimization</i> |
| 14:00 | HS3<br><i>Theory of Estimation-of-Distribution Algorithms</i>         | HS4<br><i>Large Language Models as Tools for Metaheuristic Design</i>               |                                                                                                                                          | HS3<br><i>Structural bias in optimisation algorithms</i>                                  | HS4<br><i>Transfer Learning in Evolutionary Spaces</i>                     |                                                                                                            |
| 14:30 |                                                                       |                                                                                     |                                                                                                                                          |                                                                                           |                                                                            |                                                                                                            |
| 15:00 | <b>Coffee Break</b><br>10:30 - 11:00, Break Room                      |                                                                                     |                                                                                                                                          | <b>Coffee Break</b><br>10:30 - 11:00, Break Room                                          |                                                                            |                                                                                                            |
| 15:30 | <b>Tutorial 7</b><br>15:30 - 17:00                                    | <b>Tutorial 8</b><br>15:30 - 17:00                                                  |                                                                                                                                          | <b>Tutorial 15</b><br>15:30 - 17:00                                                       | <b>Tutorial 16</b><br>15:30 - 17:00                                        |                                                                                                            |
| 16:00 | HS3<br><i>Constraint Handling in Multiobjective Optimization</i>      | HS4<br><i>Instance Space Analysis for Rigorous and Insightful Algorithm Testing</i> |                                                                                                                                          | HS3<br><i>Adversarial Optimisation through Competitive Co-evolutionary Algorithms</i>     | HS4<br><i>Robot Evolution</i>                                              |                                                                                                            |
| 16:30 |                                                                       |                                                                                     |                                                                                                                                          |                                                                                           |                                                                            |                                                                                                            |

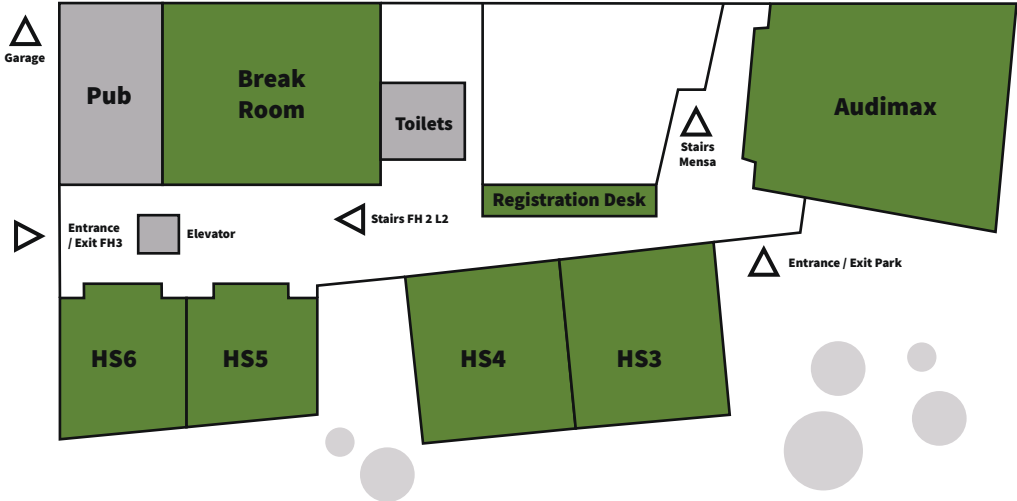
|       | Monday, 16.9.2024                                                                                                                               | Tuesday, 17.9.2024                                                                                                                               | Wednesday, 18.9.2024                                                                                                                        |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 8:00  | <b>Registratgion &amp; Info Desk</b><br>8:00 - 9:00<br>Lecture Hall Corridor                                                                    | <b>Registratgion &amp; Info Desk</b><br>8:00 - 9:00<br>Lecture Hall Corridor                                                                     | <b>Registratgion &amp; Info Desk</b><br>8:00 - 9:00<br>Lecture Hall Corridor                                                                |
| 8:30  |                                                                                                                                                 |                                                                                                                                                  |                                                                                                                                             |
| 9:00  | <b>Opening</b><br>9:00 - 9:30, Audimax                                                                                                          | <b>Keynote by Richard Küng</b><br>9:00 - 10:00<br>Audimax                                                                                        | <b>Keynote by Bernadino Romera-Paredes</b><br>9:00 - 10:00<br>Audimax                                                                       |
| 9:30  | <b>Keynote by Oliver Schütze</b><br>9:00 - 10:00<br>Audimax                                                                                     |                                                                                                                                                  |                                                                                                                                             |
| 10:00 |                                                                                                                                                 | <b>Coffee Break</b><br>10:30 - 11:00, Break Room                                                                                                 | <b>Coffee Break</b><br>10:30 - 11:00, Break Room                                                                                            |
| 10:30 | <b>Coffee Break</b><br>10:30 - 11:00, Break Room                                                                                                | <b>Poster Session 4</b><br>11:00 - 13:00<br>HS5<br><br><i>Paper IDs:</i><br>73, 125, 140, 153, 156,<br>163, 167, 185, 199, 208,<br>294, 300      | <b>Poster Session 7</b><br>11:00 - 13:00<br>HS5<br><br><i>Paper IDs:</i><br>81, 176, 179, 186, 211,<br>213, 220, 227, 230, 265,<br>279, 286 |
| 11:00 | <b>Poster Session 1</b><br>11:00 - 13:00<br>HS5<br><br><i>Paper IDs:</i><br>8, 38, 44, 62, 76,<br>94, 116, 123, 149,<br>276, 277, 296, 298      |                                                                                                                                                  |                                                                                                                                             |
| 11:30 |                                                                                                                                                 |                                                                                                                                                  |                                                                                                                                             |
| 12:00 |                                                                                                                                                 |                                                                                                                                                  |                                                                                                                                             |
| 12:30 | <b>Lunch Break</b><br>12:30 - 13:30<br>Break Room                                                                                               | <b>Lunch Break</b><br>12:30 - 13:30<br>Break Room                                                                                                | <b>Lunch Break</b><br>12:30 - 13:30<br>Break Room                                                                                           |
| 13:00 |                                                                                                                                                 |                                                                                                                                                  |                                                                                                                                             |
| 13:30 | <b>Poster Session 2</b><br>13:30 - 15:00<br>HS6<br><br><i>Paper IDs:</i><br>56, 82, 96, 99, 103,<br>105, 109, 113, 142, 143,<br>158, 204, 223   | <b>Poster Session 5</b><br>13:30 - 15:00<br>HS6<br><br><i>Paper IDs:</i><br>110, 157, 164, 165, 166,<br>171, 212, 228, 232, 239,<br>260, 274     | <b>Poster Session 8</b><br>13:30 - 15:00<br>HS6<br><br><i>Paper IDs:</i><br>9, 15, 43, 147, 226,<br>233, 234, 236, 254, 261,<br>285, 303    |
| 14:00 |                                                                                                                                                 |                                                                                                                                                  |                                                                                                                                             |
| 14:30 |                                                                                                                                                 |                                                                                                                                                  |                                                                                                                                             |
| 15:00 | <b>Coffee Break</b><br>15:00 - 15:30, Break Room                                                                                                | <b>Coffee Break</b><br>15:00 - 15:30, Break Room                                                                                                 | <b>Closing</b><br>15:00 - 15:30, Audimax                                                                                                    |
| 15:30 | <b>Poster Session 3</b><br>15:30 - 17:00<br>HS5<br><br><i>Paper IDs:</i><br>16, 67, 106, 114, 120,<br>126, 146, 152, 154, 168,<br>177, 256, 297 | <b>Poster Session 6</b><br>15:30 - 17:00<br>HS5<br><br><i>Paper IDs:</i><br>89, 115, 127, 190, 193,<br>200, 201, 207, 221, 242,<br>252, 269, 302 |                                                                                                                                             |
| 16:00 |                                                                                                                                                 |                                                                                                                                                  |                                                                                                                                             |
| 16:30 |                                                                                                                                                 |                                                                                                                                                  |                                                                                                                                             |

# Venue Map

## Softwarepark Hagenberg Center



## FH2 Level 0



# Social Events

The program includes two social events during the main conference days: A welcome reception at the Ars Electronica Center (AEC) Linz and the conference dinner at the restaurant of our local brewery "Freistädter Bier". For both events a shuttle bus service to the event venues and back to the conference venue Hagenberg and to Linz will be provided.



## Welcome Reception at Ars Electronica Center

*Monday 19:00 - 22:00, September 16, 2024*

*Ars-Electronica-Straße 1, 4040 Linz, Austria*

On Monday, we invite you to the Ars Electronica Center (AEC) Linz - museum of the future - where interactive stations, works of art, research projects, large-scale projections and laboratories are presented. After the Deep Space 8k session you'll enjoy a flying dinner at the rooftop restaurant. For any further information regarding the Ars Electronica Center, have a look at their website.

<https://ars.electronica.art/center/en/>



## Conference Dinner at Freistädter Brauhaus

*Tuesday 19:00 - 22:00, September 17, 2024*

*Brauhausstraße 2, 4240 Freistadt, Austria*

On Monday, we invite you to the Freistädter Brauhaus, where you'll have the opportunity to taste the local beer, lemonades, etc. and enjoy the cuisine of the "Mühlviertel" region, where this year's edition of PPSN is taking place. After a welcome drink at the brewery, we'll have dinner at their restaurant with regional specialities. For any further information regarding the Freistädter Brauhaus, have a look at their website.

<https://www.freistaedter-bier.at/>

## Shuttle Bus

We provide a shuttle bus service between the conference venue **FH OÖ Hagenberg**, Linz city center at **Lentos Art Museum Shipping Pier** and the social event venues **Ars Electronica Center Linz** & **Freistädter Brauhaus**.

|    |       |       |                        |                          |
|----|-------|-------|------------------------|--------------------------|
| Sa | 14.9. | 07:45 | Lentos Shipping Pier   | → FH OÖ Hagenberg        |
| Sa | 14.9. | 17:45 | FH OÖ Hagenberg        | → Lentos Shipping Pier   |
| Su | 15.9. | 07:45 | Lentos Shipping Pier   | → FH OÖ Hagenberg        |
| Su | 15.9. | 17:45 | FH OÖ Hagenberg        | → Lentos Shipping Pier   |
| Mo | 16.9. | 08:00 | Lentos Shipping Pier   | → FH OÖ Hagenberg        |
| Mo | 16.9. | 18:30 | FH OÖ Hagenberg        | → Ars Electronica Center |
| Mo | 16.9. | 22:00 | Ars Electronica Center | → FH OÖ Hagenberg        |
| Tu | 17.9. | 08:00 | Lentos Shipping Pier   | → FH OÖ Hagenberg        |
| Tu | 17.9. | 18:30 | FH OÖ Hagenberg        | → Freistädter Brauhaus   |
| Tu | 17.9. | 22:00 | Freistädter Brauhaus   | → FH OÖ Hagenberg        |
| We | 18.9. | 08:00 | Lentos Shipping Pier   | → FH OÖ Hagenberg        |
| We | 18.9. | 16:15 | FH OÖ Hagenberg        | → Lentos Shipping Pier   |

**Pfiat eich! [pfiat aɪç]** *common Austrian farewell formula*

We hope you've enjoyed your stay in Hagenberg, Upper Austria and experienced a remarkable PPSN 2024!

We look forward to seeing you again in Hagenberg!



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